

EMMANUEL ARO

Full-Stack & Systems Software Engineer | Distributed Systems & Applied AI

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EXECUTIVE TECHNICAL PROFILE

Full-Stack & Systems Software Engineer with 6+ years of professional engineering experience architecting distributed platforms, real-time systems, and production-grade applied AI pipelines. Proven track record of shipping end-to-end solutions using Python (Django Ninja), TypeScript (Next.js App Router), PostgreSQL, and Docker across 7 companies globally. Adept at designing defensible API contracts, orchestrating GPU serverless infrastructure for ML workloads, and optimizing database layers to ensure high availability, low latency, and massive concurrency under load.

TECHNICAL ARSENAL

- **Languages:** Python, TypeScript, JavaScript (ES6+), PHP, SQL, HTML5/CSS3
- **Frameworks & Full-Stack:** Next.js (App Router), React, Django / Django Ninja, Laravel, FastAPI, Node.js / Express.js
- **Distributed Systems & APIs:** RESTful APIs, GraphQL, WebSockets, Celery, Redis, Task Queues
- **Applied AI & Computer Vision:** Agentic RAG (LlamaIndex, LangGraph), Ollama, OpenRouter, WhisperX, LexNLP, OpenCV, Scikit-learn
- **Databases & Storage:** PostgreSQL (pgvector, RLS), MongoDB, Supabase, MySQL, Redis, Schema Modeling & Query Optimization
- **Cloud, DevOps & Tooling:** Docker, Docker Compose, AWS (EC2, S3, RDS), Modal (Serverless GPU), Linux/Bash, GitHub Actions CI/CD, Postman

PROFESSIONAL EXPERIENCE

LunarTech

Software Engineer – AI Systems & Platform Engineering

May 2026 – Present | Austin, TX (Remote)

- Engineered Agentic RAG pipelines (LangGraph, LlamaIndex) for Rosendahl, a self-hosted legal AI workspace, delivering fully evidence-linked and formally verified outputs across 121+ active ECHR case workspaces.
- Designed legal document processing APIs using LexNLP and spaCy for automated entity extraction and risk scoring, enforcing a 4-tier confidential AI model routing strategy (OpenRouter, Claude, local Ollama) to maintain strict data privacy.
- Architected Dark Phoenix, an end-to-end GPU video pipeline on Modal (TalkNet, WhisperX, Gemini AI, ffmpeg) for automated podcast clipping, YouTube ingestion, and provider-agnostic S3 storage.
- Executed comprehensive performance QA on Octavia, an AI dubbing platform (60+ languages, 99% lip-sync accuracy), implementing critical database optimization techniques that eliminated bottlenecks and enabled seamless scaling for over 10,000 users worldwide.

Boxonia Blueprint

Software Engineer (Platform & Ecosystem Architecture)

Oct 2024 – Present | Lagos, NG (Contract / Concurrent)

- Conceived and engineered a greenfield digital platform for a 360° film production and talent management studio, establishing the company's first robust online presence and distributed story workflow engine.
- Architected a full Python-based platform ecosystem, consolidating workflows across 5+ distributed production teams to scale story development, talent directory management, and project portfolio showcasing.
- Developed an automated talent-booking and cross-border coordination system, enabling zero-friction client onboarding and reducing scheduling operational overhead by 40%.
- Scaled relational schemas and secured RESTful APIs to ensure high-availability data synchronization between international production units under strict SLA requirements.

Tech4mation (NativeTalk)

Software Developer, Platform Engineering

Feb 2026 – Jun 2026 | Lagos, NG

- Led full-stack architecture for NativeTalk CRM (Next.js App Router, Django Ninja, PostgreSQL), driving aggressive bug-triage cycles that resolved critical customer-facing issues and directly increased company revenue.
- Architected TaskForge Pro, a high-throughput internal team-management platform with decoupled microservice boundaries and strictly typed API contracts.
- Standardized engineering environments by implementing containerized Docker workflows, successfully reducing new-developer onboarding time from 3 days to under 2 hours.

Ornament & Crime

Frontend Engineer (Shopify Developer)

Aug 2024 – Mar 2026 | Birmingham, AL, US

- Architected the migration of a legacy Shopify Plus storefront to a componentized Liquid 2.0 and Tailwind CSS system, reducing page load times by 48% and driving a 31% uplift in checkout conversion.
- Engineered custom Shopify app integrations (inventory sync, loyalty tiers, fulfillment webhooks) via the Shopify Admin GraphQL API, eliminating manual operations and saving 20+ staff hours per week.
- Led the brand's migration to a headless-compatible architecture, setting the engineering foundation for future Next.js storefront decoupling with zero downtime.

Bincom Dev Center

Full Stack Developer

Sep 2025 – Jan 2026 | Lagos, NG

- Architected SEO-optimized frontend interfaces using Next.js (App Router) and React, leveraging Server Components and advanced data-fetching patterns to reduce First Contentful Paint (FCP) by 35%.
- Overhauled relational MySQL schemas and refactored critical SQL queries, eliminating latency spikes and accelerating complex data retrieval by 20% under peak load.
- Mentored junior engineers in TypeScript best practices, modular component design, and strict Git workflows, measurably elevating team delivery velocity.

CyberBuddies

Frontend Engineer

Apr 2024 – Aug 2024 | Ibadan, NG

- Engineered high-quality frontend UI components for an AI-powered Business Operating System™, establishing core visual language patterns that directly elevated user satisfaction and drove new B2B revenue growth.
- Supported the migration of production workloads to AWS infrastructure, implementing optimization strategies that reduced cloud operating costs by 18% while maintaining 99.9% uptime SLAs.

SafeSpace.org Association

Operations QA & Automation Engineer

May 2023 – Feb 2024 | California, US (Remote)

- Deployed Python/Django automated review pipelines, replacing manual verification loops to achieve a 30% throughput gain and a 50% reduction in data-entry errors.
- Instituted rigorous automated testing protocols within containerized Docker environments to establish company-wide benchmarks for system integrity.

Fiatci.io

Lead Full Stack Engineer

Sep 2022 – Mar 2023 | Brussels, BE (Remote)

- Rapidly promoted to Lead Full Stack Engineer based on consistent technical delivery, taking full ownership of a 7-person cross-functional engineering team.
- Directed the end-to-end architecture, development, and production deployment of the company's flagship platform, culminating in the first successful commercial product launch in company history.
- Established comprehensive technical direction across frontend, backend, and DevOps infrastructure, implementing rigorous code review standards, CI/CD release strategies, and zero-downtime production cutover protocols.

FLAGSHIP SYSTEMS & APPLIED AI PROJECTS

- **Geod (Sovereign AI Workspace):** Architected a unified real-time collaboration workspace featuring WebRTC audio/video, multi-modal OCR, a sandboxed code runtime, and decoupled vector/queue pipelines with zero vendor lock-in.
- **Rosendahl (Legal AI Engine):** Engineered a legal document processing platform utilizing LangGraph and LlamaIndex for multi-tier confidential routing and LexNLP for automated entity extraction.
- **Dark Phoenix (GPU Video Ingestion Pipeline):** Developed an end-to-end automated podcast clipping pipeline deployed on Modal serverless infrastructure, integrating WhisperX, TalkNet, and Gemini AI.
- **Enterprise Malware Detection Engine:** Built a production-grade supervised ML classifier for real-time endpoint threat detection, incorporating automated feature extraction and continuous retraining pipelines.
- **Clinical Oncology Risk Prediction Model:** Developed an explainable, high-accuracy diagnostic classification model using Scikit-learn and Pandas, prioritizing reproducible feature engineering and interpretable outputs for clinical decision support.

EDUCATION & CREDENTIALS

- B.Sc. in Computer Science — Federal University Oye-Ekiti, Nigeria (2025)
- Full Stack Engineer Certification — Google Developer Student Clubs (2024)
- Advanced Software Engineering Certification — GB-Tech Learning Centre, Nigeria (2023)